



Adapting a Latent Audio Diffusion Model to Historical Guqin Recordings: A Listening-Driven Case Study

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RESEARCH ARTICLE

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ABSTRACT

We report a small-data case study in adapting a pretrained latent audio diffusion model to the guqin, the seven-string Chinese zither, aiming at an “AI radio” that plays guqin-style music without end. From a library of historical recordings we curate 412 solo performances (42.5 h, 61 performers) and split them by composition. We fine-tune a rank-16 DoRA adapter on Stable Audio 3 Medium using its continuous latents, masking weighted towards continuation, captions that combine researched notes on each piece with mood tags and an automatically estimated pentatonic mode, and random-length crops, stopping when held-out loss stops improving. Seven blind listening studies by one expert listener guided every decision. The final adapter was rated highest for continuing unseen pieces (3.9/5, against 3.4 for the best earlier adapter) and 4.6/5 for generating from free-written scene descriptions. Negative results are equally informative: a from-scratch autoregressive model over the same latents produced no recognisable timbre, a signal-level friction-noise measure correlated with the listener’s complaints in the wrong direction, and a pentatonic-fit measure tracked ratings overall but barely within a group of candidates. Chaining continuations for long playback exposed a silent tail on every generated clip and a loudness feedback loop, both with simple fixes. With one listener and at most ten clips per condition, no paired difference is statistically significant; we present an exploratory record of what helped, what did not, and why.

KEYWORDS:

guqin, music generation, latent diffusion, low-resource fine-tuning, listening evaluation

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1 INTRODUCTION

Text-to-music and audio continuation models are trained mostly on Western popular and library music. The guqin is a poor fit for that prior: its repertoire is largely pentatonic, its expressive vocabulary lives in left-hand slides and vibrato (吟猱绰注) whose passing semitones are idiomatic only in specific places, and the best-known performances survive mainly as mid-twentieth-century recordings with tape hiss and limited bandwidth. The amount of usable solo audio is also small: tens of hours, not thousands.

Our practical goal is an “AI radio”: a system that keeps producing guqin-like music for hours, either continuing from a recorded opening or following a short textual description of a scene. This article documents how we got from a general-purpose model to an adapter that a guqin-literate listener rates as convincing, and records the dead ends along the way, because in a domain this small the negative results are a large part of what was learned.

Our contributions are:

- A curation and evaluation protocol for a small historical corpus: listening screening of every recording, composition-level splits so that test pieces are never heard in another performer’s version, and blind, shuffled listening with free-text comments (Sections 3 and 5).
- A fine-tuning recipe for Stable Audio 3 Medium (Evans et al., 2026) on 42 h of audio: DoRA (Liu et al., 2024) on the base model with inference on the distilled model, masking weighted towards prefix continuation, captions that mix researched notes on each composition with mood tags and an estimated pentatonic mode, random-length crops, and early stopping on held-out loss (Section 4).
- Seven listening studies and the decisions they drove, including clear negative results for a from-scratch latent language model and for two automatic quality measures (Section 6).
- Two practical failure modes of chained continuation for long playback, with fixes (Section 7).

2 RELATED WORK

2.1 LATENT AUDIO DIFFUSION

Stable Audio (Evans et al., 2024a) generates variable-length audio by diffusion in the latent space of an autoencoder, conditioned on text and timing; Stable Audio Open (Evans et al., 2024c) and Stable Audio 3 (Evans et al., 2026) extend this with open weights, inpainting and continuation. Work on long-form generation with latent diffusion (Evans et al., 2024b) shows that the training window length matters for musical structure. We build

directly on Stable Audio 3 Medium and its continuous SAME-L autoencoder.

2.2 AUTOREGRESSIVE MUSIC MODELS

Token-based language models such as MusicGen (Copet et al., 2023) and streaming systems such as Magenta RealTime (Lyria Team et al., 2025) generate indefinitely by construction. Continuous-latent autoregression with a diffusion or flow head (Li et al., 2024) avoids quantisation, and noise augmentation of the context reduces error accumulation (Pasini et al., 2024). We test a small model of this kind trained from scratch (Section 6.4).

2.3 ADAPTATION FROM SMALL DATA

LoRA (Hu et al., 2021) and DoRA (Liu et al., 2024) adapt large models with few trainable parameters. For data-constrained training, repeating data for a few epochs is nearly as useful as fresh data (Muennighoff et al., 2023), which motivates our multi-pass schedule with early stopping.

2.4 GUQIN AND EVALUATION

Prior guqin MIR work analyses mode and playing technique from historical recordings (Huang et al., 2020). For generated music, replication of training data is a central concern (Batlle-Roca et al., 2024); we keep evaluation pieces out of training at the composition level.

3 DATA

3.1 SOURCE AND SCREENING

The source is a private library of 638 historical guqin recordings (44.1 kHz, 16-bit stereo CD transfers). Every candidate was screened by the author, a guqin-literate listener, through a whole-track listening page; recordings were excluded for audible noise floors, ensembles or non-guqin instruments, and voice. The result is **412 solo recordings by 61 performers, 42.5 h**, with 108 distinct titles after normalising numbered variants (a few titles are alternative names of the same piece). Performer coverage is uneven: the five most represented performers account for about 40% of the duration.

3.2 SPLITS

We split by composition rather than by performer, so that no version of a test composition is heard in training: 329 recordings for training, 42 for validation and 41 for test. Each recording is cut into aligned windows of the model’s native length (59.9 s) that together cover the whole performance, giving 2,764 windows (2,217/276/271); nine windows of pure digital silence are dropped. The final run trains on the training and test recordings together (370 recordings) and keeps the 42 validation recordings out for early stopping and listening.

3.3 CAPTIONS

For each composition we compiled a short narrative from published notes on the piece (e.g. *Liushui*, Boya and his true listener Ziqi; *Guangling San*, Nie Zheng's revenge and Ji Kang's last performance), assigned two to five tags from a closed vocabulary of 37 moods and scenes (e.g. *calm*, *lonely*, *vast*, *reclusive*, *autumn*, *river*, *frontier desert*), and recorded our confidence: 55 notes are well sourced, 31 partly, and 22 inferred from the title. A pentatonic mode (the *gong* pitch and its five notes) is estimated automatically for every window from a tuning-corrected pitch-class histogram. Captions are written in English with the Chinese text appended and never exceed 160 tokens of the 256-token encoder limit.

4 METHOD

4.1 MODEL

Stable Audio 3 Medium (Evans et al., 2026) is a diffusion transformer over SAME-L latents (256 dimensions at about 10.8 frames per second) conditioned on T5Gemma text embeddings and on an inpainting mask. Following the released recipe, we train a rank-16 DoRA adapter (21.6M trainable parameters) on the *base* model and apply it to the step-distilled *Medium* model at inference (8 sampling steps). All latents are pre-encoded once.

4.2 OBJECTIVE, MASKING AND TEXT

Training uses the model's rectified-flow objective (Liu et al., 2022) with its inpainting conditioning. We change the masking mixture from the default 10/80/10 (random spans / fully masked / prefix continuation) to 10/30/60, so that most training examples resemble our main use, continuing a given opening. Each window receives one caption style by a stable hash: 40% tags and mode, 40% narrative, tags and mode, 20% narrative only; the trainer additionally drops 10% of captions.

4.3 CROPS AND SCHEDULE

Each sample keeps a random span of 20–60 s at a random start inside its window, padded with the silence latent and with the timing condition set to the kept length. One training pass is 2,500 steps at batch size 1 (about one epoch), learning rate 10^{-4} , AdamW. After each pass we compute the model's validation loss at five fixed noise levels on all 276 validation windows and stop when three consecutive passes fail to improve on the best. Checkpoints every 500 steps make the run resumable after interruptions.

4.4 INFERENCE

Continuation keeps a 10 s opening and generates the next 25 s; text-only generation produces 35 s from a description. For long playback we chain continuations (Section 7).

5 EVALUATION PROTOCOL

All listening was done by the author, who has extensive experience with the guqin repertoire, on local web pages in a fixed format: candidates for the same opening and seed are shuffled and unlabelled; each clip gets an overall score from 1 to 5, optional issue tags (e.g. *too many semi-tones*, *sawing/metallic noise*, *not guqin-like*, *no melody*, *abrupt seam*) and free-text comments; the mapping is revealed only after rating. Openings come from compositions the compared models did not train on, with each study using different performers and pieces where possible. For paired comparisons we report exact two-sided sign-flip permutation tests over openings and percentile bootstrap intervals; with eight or fewer pairs these tests have little power and we treat them as descriptive.

6 RESULTS

Table 1 summarises the seven studies. Scores are means over clips; n is clips per condition.

6.1 SEED AND OPENING DOMINATE

In R1 the two seeds differed by 1.6 points on average (2.17 against 3.75), more than any model difference; in R2 the mean per opening-and-seed group ranged from 1.0 to 4.75 while inference settings moved the mean by at most 0.1 (except the base model). The same semi-tone often appeared at the same second in all candidates sharing an opening and seed. This variance motivates generating several takes and selecting, and it is why per-condition means over a handful of clips must be read cautiously.

6.2 INFERENCE SETTINGS AND TEXT CONDITIONING

Lowering adapter strength or restricting the adapter to early denoising steps made no difference. Using the base model with 50 steps, the configuration matching training, was clearly worse (2.70, five scores of 1, comments about extra instruments), confirming that the distilled model is the better renderer. Training with mood tags left continuation unchanged (matching and deliberately opposite tags both scored 4.0: the opening dominates) but improved generation from text alone from 2.0 to 3.8 ($p = 0.06$, $n = 5$).

6.3 TRAINING LENGTH AND EARLY STOPPING

On the 329-recording split, held-out loss decreased monotonically over three passes (0.5387 for the base model, then 0.5351, 0.5343 and 0.5337) with no sign of overfitting; captions and mode text did not change the loss. In the final run (Figure 1) the loss fell from 0.6020 to a minimum of 0.5950 at pass 14 and did not improve over

Study	Condition	<i>n</i>	Mean score
R1 masking	default mask / continuation mask / 412 recordings, continuation mask	8	2.88 / 2.62 / 3.38
R2 inference	adapter strength 1.0 / 0.7 / adapter on early steps only	9–10	3.56 / 3.60 / 3.67
	base model with 50 steps / earlier 222-clip adapter	10	2.70 / 3.10
R3 captions	continuation: no text in training / no prompt / matching / opposite tags	9	3.89 / 3.56 / 4.00 / 4.00
	text only: adapter trained without / with captions	5	2.00 / 3.80
R4 latent LM	from-scratch latent LM / SA3 adapter	11 / 6	1.00 / 2.00
R5 training length	1 pass / 3 passes + tags / 3 passes + tags + mode	8	3.75 / 3.75 / 4.12
R6 narratives	pass 2 / adapter C from R5	7 / 8	3.00 / 3.50
R7 final	continuation: pass 14 / pass 2 / adapter C	8	3.88 / 2.75 / 3.38
	scene descriptions: pass 14 / pass 2	5	4.60 / 3.60

Table 1 Overview of the listening studies (scores 1–5, single expert listener). In R1 the 412-recording adapter had seen the validation compositions; all other comparisons use openings unseen by every compared model, except R2 (see Section 8).

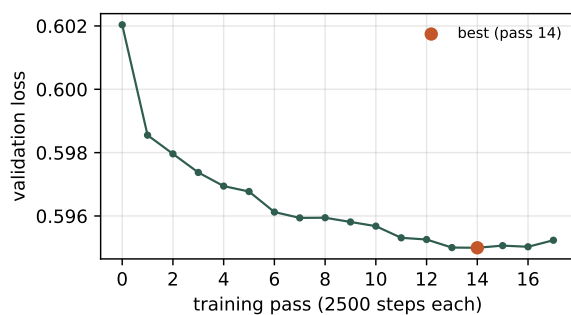


Figure 1 Validation loss per training pass of the final run (276 held-out windows, five fixed noise levels).

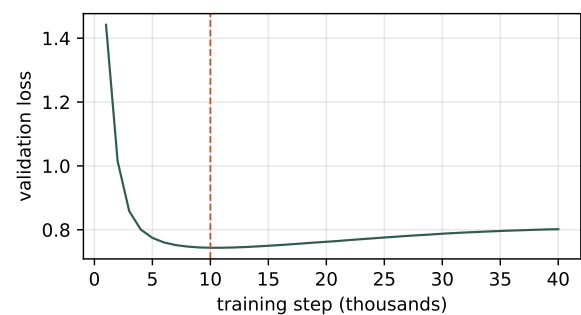


Figure 2 Validation loss of the from-scratch latent language model; it rises after 10k steps (dashed line).

the next three passes, triggering early stopping. The pass-14 adapter improved continuation over the pass-2 checkpoint of the same run (3.88 against 2.75, $p = 0.16$) and over adapter C (3.38, $p = 0.63$), and received no “sawing” tags against three for pass 2. Bootstrap 95% intervals are wide ([2.75, 4.75] for pass 14). For scene descriptions written after training, pass 14 scored 4.6 against 3.6 ($p = 0.31$); one take was singled out by the listener as exceptional.

6.4 NEGATIVE RESULT: A FROM-SCRATCH LATENT LANGUAGE MODEL

We trained a causal transformer (8 layers, width 512) with a flow-matching MLP head that predicts the next SAME-L frame, with context noise augmentation (Pasini et al., 2024) (72 M parameters, 1.42 M training frames). Validation loss reached its minimum at 10k steps and rose by 8% by 40k steps (Figure 2). Every one of its 11 clips received the lowest score, including endless and unconditioned generation, with comments such as “no timbre at all”: independently sampled frames left the region that the decoder renders as guqin. The same openings continued by the SA3 adapter scored 2.0. At this data scale, a frame-level autoregressive model over continuous latents is not competitive with adapting a large pre-trained generator.

6.5 AUTOMATIC QUALITY MEASURES

The listener’s most frequent complaint was “sawing”: left-hand friction noise that is idiomatic in moderation but grating when overused. A measure of noisy high-band frames correlated with the score in the wrong direction (Spearman $\rho = +0.39$ over 48 clips), and the nine clips tagged as sawing had lower noisy-frame fractions than the rest (0.0006 against 0.0017).

The fraction of tonal energy on the best-fitting pentatonic set tracked scores over 124 rated clips ($\rho = 0.52$, positive in each study) but only weakly among candidates for the same opening ($\rho = 0.31$ after removing group means). Choosing the best-fitting candidate raised the mean from 3.41 to 3.52; discarding the worst raised it to 3.58. Restricting the measure to pitches at note onsets, motivated by the listener’s observation that idiomatic semitones occur in slides rather than at the pluck, was weaker still ($\rho = 0.19$ within groups), although generated clips did place more onsets outside the mode than real recordings (33% against a median of 22%; 39% in clips flagged for semitones). The measure also disagreed with R5, where it predicted that longer training worsened modal fit while the listener preferred the longer-trained model.

6.6 RATER CONSISTENCY

Fifteen clips from R6 were rated again, unknowingly, in R7. Six scores were identical, twelve within one point, and

the mean absolute difference was 0.8. A clip chosen for the showcase in one session received a 1 in the next when heard next to an alternative. Context effects of this size are comparable to the differences between our models and are the main reason we do not claim significance.

7 LONG PLAYBACK BY CHAINED CONTINUATION

To extend the favourite 35 s clip, we repeatedly fed its last 15 s as an opening and appended 30 s of new material. Two problems appeared.

7.1 SILENT TAILS

Every generated clip ends with about 0.5 s of digital silence (−87 dBFS) as the requested duration runs out. When that tail is part of the next opening, the model treats the piece as ended and inserts a further two seconds of silence at the seam. Trimming the source's last 1.5 s and over-generating each segment by 3 s, then discarding the tail, removed the gaps.

7.2 LOUDNESS FEEDBACK

Each segment came out slightly louder than its opening, and that segment became the next opening: over six rounds the level rose from −10 to −5.5 dBFS with up to 10% of samples clipped. Feeding every opening at a fixed level (−16 dBFS) and matching each new segment to the level of the 15 s before it kept the 3.5-minute result between −13.8 and −16.7 dBFS per 30 s with no clipping. Of two locked chains, the listener accepted one and rejected the other for a sudden breakdown after 82 s, a reminder that long-range stability is not solved.

8 LIMITATIONS

All ratings come from one listener, who is also the author and designed the systems; the studies are small and were run sequentially as development proceeded, so later studies are not independent of earlier decisions. In R2 the openings came from a performer who, contrary to the original plan, had been included in the 412-recording training set. The narratives and tags are condensed readings of traditional notes on each piece and simplify a rich interpretive tradition; 22 of the 108 titles are inferred from the title alone. Multi-listener studies, objective measures validated against them, and checks for replication of training material (Batlle-Roca et al., 2024) are needed before stronger claims.

9 CONCLUSION

A pretrained latent diffusion model, a few tens of hours of carefully screened historical recordings, a small adapter

and a single attentive listener were enough to produce guqin continuations and scene-driven pieces that the listener rated highly, while training a generator from scratch on the same data failed completely. The strongest effects came not from architecture but from data curation, held-out early stopping, and simple engineering of the generation loop. The next steps are multi-listener evaluation, preference-based fine-tuning on the collected ratings (Wallace et al., 2023), and stable long-form generation for continuous playback.

10 REPRODUCIBILITY

Code for corpus preparation, training, generation, listening pages and analysis, together with every listening rating and validation log and the script that recomputes all numbers and figures in this article, is available at <https://github.com/HuanchenCai/guqin-generation> (tag freeze-2026-10-05, which also records the SHA-256 hashes of all checkpoints used). The training recordings are commercially published historical performances protected by copyright and cannot be redistributed, and for the same reason we do not release audio generated from them or the adapter weights. The base model is used under Stability AI's community license.

11 ETHICS AND CONSENT

The listening studies involved no participants other than the author. No personal data were collected.

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COMPETING INTERESTS

The author has no competing interests to declare.

AUTHORS' CONTRIBUTIONS

HC conceived the project and its goal of an endless guqin "AI radio", and directed every stage of the exploration. HC screened the library by listening to whole recordings, decided which performers and pieces to keep or exclude (including recordings with noise floors, ensembles, applause or voice), and noticed that a DVD collection might duplicate the CD material. HC required that splits be made by composition rather than by performer, that evaluation openings never repeat a piece or performer, and that validation recordings stay out of training so that overfitting could be measured. HC proposed moving from

text-free continuation to mood labels derived from piece titles, and then to captions telling the story behind each title; proposed the move towards a language-model-like generator over latents; observed that idiomatic guqin semitones occur in slides rather than at the pluck, which shaped the onset-based measure; proposed random crop lengths in addition to random start points; asked for training to continue until validation loss genuinely stopped improving, and for the run to be resumable after power cuts. HC performed all listening studies and wrote all ratings and comments, identified the silent seams and the loudness drift in chained continuation, and selected the takes used for demonstration.

USE OF AI TOOLS

AI coding assistants (OpenAI Codex and Anthropic Claude) implemented and ran the code for data processing, training, generation, listening pages and analysis under the author's direction, searched for and summarised background sources on the pieces, and drafted parts of this manuscript. All research decisions and all listening judgements are the author's, who reviewed the results and the text and takes responsibility for the content.

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